

Abstract

Bangladesh faces a high rate of road accidents, especially involving motorcycles, due to dense traffic and reckless riding. This study develops an AI-based traffic monitoring system within an Intelligent Transportation System (ITS) framework to detect violations such as wrong-lane driving, overspeed, helmet absence, and multi-rider scenarios.

The system uses YOLOv11 for real-time detection of vehicles and violations from live video, allowing rapid identification without extra hardware. Detected license plate regions are processed by a custom CRNN trained on Bangla plates, which extracts and recognizes the alphanumeric sequence using token-based decoding and pattern matching.

An automated alert mechanism sends SMS notifications to drivers and authorities when violations are detected. The system was tested on real-world videos from Chittagong. Results show high effectiveness, with license plate detection at 96%, helmet detection at 84%, and no-helmet detection at 76%. Plate recognition precision was 94%.

These results demonstrate that combining YOLOv11 for object detection and CRNN for license plate recognition provides a scalable, real-time solution for traffic enforcement. The framework is cost-effective, adaptable to urban conditions, and supports law enforcement in preventing accidents. By automatically identifying risky behavior and alerting drivers, the system contributes to safer roads and improved traffic management in Bangladesh.

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Nomenclature

Symbols	Description
V_{ID}	Speed of the vehicle with tracking ID
X_1	Distance from camera to the first virtual reference line
X_2	Distance from camera to the second virtual reference line
T	Time taken by a vehicle to travel between two references
IoU	Intersection over Union between two bounding boxes
B_p	Bounding box of a detected person
B_v	Bounding box of a detected vehicle
$\mathcal{L}_{CTC}$	Connectionist Temporal Classification loss
x_t	Feature representation at time step t
y	Ground truth label sequence
π	Possible alignment path for CTC decoding
$\mathcal{B}_{-1}$	Mapping function
$D(i, j)$	Levenshtein edit distance
s_i	i^{th} character of string s
t_j	j^{th} character of string t

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
ANN	Artificial Neural Network
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
BiLSTM	Bidirectional Long Short-Term Memory
CRNN	Convolutional Recurrent Neural Network
CTC	Connectionist Temporal Classification
RF	Random Forest
ResNet	Residual Network
YOLO	You Only Look Once
R-CNN	Region-based Convolutional Neural Network
SSD	Single Shot Detector
ITS	Intelligent Transportation System
OCR	Optical Character Recognition
GPU	Graphics Processing Unit
IoU	Intersection over Union
mAP	mean Average Precision
FPS	Frames Per Second

Glossary of Terms

Term	Definition
Object Detection	The task of identifying and localizing objects within an image or video frame using bounding boxes
Bounding Box	A rectangular box that defines the spatial location of a detected object in an image
Confidence Score	Probability shown by a model indicating how certain it is that a detected object belongs to a given class
Precision	The proportion of correctly predicted positive detections out of all positive detections made
Recall	The proportion of correctly predicted positive detections out of all actual positive instances in the dataset
F1 Score	Harmonic mean of precision and recall
Confusion Matrix	A table used to evaluate classification performance by comparing predicted and true class labels
Tracking ID	A unique identifier assigned to a detected object to maintain its identity across consecutive video frames

Chapter 1: INTRODUCTION

1.1 BACKGROUND

Road traffic accidents remain a serious public health problem worldwide, resulting in approximately 1.19 million deaths each year [1]. Low- and middleincome countries carry a disproportionately large share of these fatalities, even though they own less than half of the world's registered vehicles [1]. Urban areas in these countries experience higher accident rates due to dense traffic, poor infrastructure and limited enforcement of traffic rules. Among the most vulnerable road users are motorcyclists, who face a higher risk of serious injury or death, particularly when riding without helmets [2] or violating traffic regulations such as overspeed, wrong-lane driving, ignoring traffic signals. Studies indicate that helmet usage can reduce the risk of death among motorcyclists by more than 6 times and the risk of brain injury by up to 74%. [1].

Traditional traffic monitoring methods rely heavily on manual observation by traffic personnel, which is time-consuming, inconsistent and often ineffective in crowded urban areas. Manual enforcement is also limited by human error, fatigue and the sheer volume of vehicles in metropolitan regions. Recent advances in computer vision and deep learning have enabled the development of real-time systems capable of detecting vehicles, monitoring rider behaviour and identifying rule violations automatically. Object detection models such as YOLO (You Only Look Once) [3] have been widely used for detecting multiple objects in video streams with both speed and accuracy. Leveraging these techniques, traffic authorities can improve road safety by enabling faster detection, reducing human intervention and issuing real-time alerts to drivers when violations occur.

1.2 CONTEXT

Due to the complexity of urban traffic, traditional traffic monitoring techniques frequently fall short, delaying enforcement and decreasing road safety. Alternative object detection models or previous YOLO variations have been used in a number of studies to suggest traffic monitoring solutions. However, complex traffic conditions cause performance reduction in many of the current systems.

Automated traffic monitoring systems that can identify moving offenses instantly have been made possible by recent advancements in computer vision and deep learning. Artificial intelligence methods can process large image datasets efficiently and with high accuracy. Popular single-stage detection methods include SSD(Single Shot Detector), RetinaNet [4] and YOLO(You Only Look Once) [3] as they can perform detection using a single neural network. While two-stage detectors achieve high accuracy, they are often too slow for realtime applications [5]. In contrast, single-stage detectors aim to balance speed and reasonable accuracy [5]. Among the object detection algorithms, YOLO have become leading frameworks for applications requiring both speed and accuracy [3]. These models can identify and locate multiple objects within an image or video frame, making them well suited for real-time traffic enforcement.

This paper employs YOLOv11, the latest high-performing version of the YOLO architecture, for real-time detection of traffic safety issues. The proposed system processes live or recorded video streams, identifies helmet, speed, lane and detect if traffic rule is violated or not. If violation occur by a vehicle, license plate of that vehicle is recognized and instant alert is issued to the driver. This approach enhances automated traffic enforcement and road safety.

This system integrates helmet detection, license plate recognition, wronglane detection and speed monitoring— to detect multiple traffic violations.

1.3 AIMS AND OBJECTIVES

The primary aim of this research is to design and implement an AI-based traffic monitoring system capable of detecting traffic rule violations in real time. The system is intended to improve road safety by providing automated, accurate and timely feedback to road users. Our key objectives were:

- **Detection of Key Objects:** Create models that use the YOLO object identification technique to identify cars, helmets, and license plates from video feeds. Assure reliable detection in a variety of traffic, lighting, and angle situations.
- **Identification of Traffic Violations:** Implement algorithms to recognize rule violations such as overspeed, wrong-lane driving, and absence of helmets. Enable the system to process these detections in real time to support immediate action.
- **Automated Alert System:** Design a notification mechanism to send SMS alerts to drivers whenever a violation is detected. Ensure the alert system is fast, reliable, and scalable for urban traffic networks.
- **Performance Evaluation:** Analyze the system's processing speed, false positive and false negative rates, and detection accuracy. Achieve a balance between real-time performance and detection reliability by optimizing the models and algorithms.
- **System Usability and Practicality:** Develop a user-friendly interface that allows traffic authorities to monitor traffic via image, video or live streams. Ensure the system can integrate with existing traffic management infrastructure and provide actionable insights to reduce accidents and enforce compliance.

1.4 SIGNIFICANCE, SCOPE AND DEFINITIONS

This research has utmost significance as motorcycles comprise a large and active portion of transportation in Bangladesh. They are also involved in a significant number of fatal road accidents (28.8 deaths per 10,000 registered motorcycles) throughout the country due to severe traffic rule violations. The proposed system introduces an AI-based monitoring system that automatically detects motorcycles, helmets, the number of riders, speeding, wrong-way movement, and license plates. It automatically extracts and detects Bangla license plates and performs the other aforementioned functions using YOLOv11.

The system contributes to the advancement of AI-based traffic monitoring by enabling real-time surveillance of motorcycles on highways. By utilizing video data from highway CCTV cameras, the system continuously monitors motorcycle movements and identifies violations such as riding without a helmet, carrying more than two persons, wrong-way movement, and overspeed. Upon detecting a violation, the system automatically extracts the license plate of the offending motorcycle and generates digital evidence in the form of images and video frames. It then extracts the characters from the license plate and alerts both the user and the relevant authorities. By reducing the dependency on manual monitoring, the system can simultaneously detect multiple motorcycles that violate traffic rules. This enhances enforcement efficiency and enables authorities to identify a greater number of traffic violations in real time.

Although the system can identify all types of vehicles, its primary target is to ensure the safety of motorcycle riders. Traffic signal violation detection is not included in the current system. The system was mainly tested on motorcycles in Bangladesh. It cannot detect license plates of all languages, as it is designed primarily for Bangla license plates. Furthermore, the detection system requires a minimum level of lighting to operate effectively.

1.5 THESIS OUTLINE

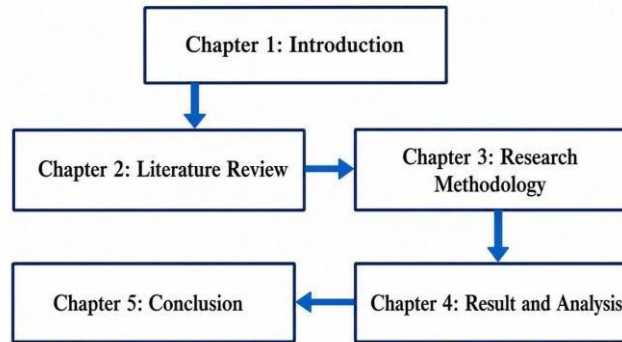


Fig. 1.1 Structure of Thesis

1st Chapter highlights the need for an Artificial Intelligence (AI)-based Intelligent Transportation System (ITS) to address major causes of road accidents in the country, such as overspeed and wrong-direction movement. It presents the background, significance and goals of the study.

2nd Chapter reviews previous studies and existing traffic monitoring technologies. It identifies the limitations of current systems in terms of real-time operation, automation and scalability, especially for developing countries like Bangladesh.

3rd Chapter describes the methodology and tools used to develop the proposed system. It explains the use of the YOLOv11 object detection model, license plate recognition and automated alert generation for detecting motorcycle traffic violations.

4th Chapter shows the obtained result from the proposed scheme. It demonstrates the system's ability to accurately detect violations, recognize license plates and generate alerts for users and authorities.

Finally, 5th Chapter discusses how the system performed, compares it with traditional approaches, also highlights its potential to improve traffic safety through intelligent and automated monitoring.

Chapter 2: LITERATURE REVIEW

2.1 GENERAL

Automated detection of traffic violations has received increasing attention in recent years, largely due to the limitations of manual enforcement in growing urban areas. With the advancement of deep learning, convolutional neural networks (CNNs) and object detection frameworks have been adopted to create more effective and scalable systems.

Franklin et al. [6] developed a traffic signal violation detection system using YOLOv3 to monitor signal jumps, vehicle speed, and vehicle count, achieving 97.67% accuracy for vehicle counting and 89.24% for speed violation detection . This work demonstrated the potential of single-stage detectors for traffic enforcement tasks.

Pavithra et al. [7] proposed an end-to-end framework for detecting helmet and crosswalk violations, combining YOLO for vehicle detection, Mask R-CNN for segmentation and OCR for license plate recognition to notify violators . This pipeline is a bit similar to the one used in the present study. More recent systems, such as Safe Streets, include detection of triple riding, helmet usage, and license plate visibility using live roadside video feeds [7]. Similarly, AI-enabled real-time enforcement platforms integrate automatic number plate recognition, helmet detection and insurance compliance verification into a unified smart city framework [8].

2.2 HISTORICAL BACKGROUND

Road accidents have been the most prominent cause of unwanted deaths of Bangladeshi citizens throughout the years. Surveys show bikers are the worst victims due to unplanned traffic scenarios, lack of strict implementation of traffic rules, and inadequacy of digitization in this sector. “In September, a total of 417 people were killed and 682 injured in 446 road accidents across the country. Among them, 143 people lost their lives in 151 motorcycle accidents, making motorcycles responsible for 33.87 per cent of accidents and 34.29 per cent of fatalities”[9].

“Road accidents claimed at least 41 lives and injured numerous others across the country during the Eid holidays since Monday,” during the Eid Festival 2025[10] . “Bangladesh witnessed 7,294 road accident deaths in 2024: Report.

Among the deceased, 893 (12.24 per cent) were women, and 1,152 (15.79 per cent) were children. Motorcycles accounted for a significant portion of road accidents, with 2,609 fatalities (35.76 per cent) from 2,761 motorcycle accidents” [11]. Bangladesh has a high rate of road fatalities (about 67) per thousand kilometers of road. “In Bangladesh, currently, only 2% of roads have a 3-star or better safety rating for pedestrians and 5% for cyclists, lagging significantly behind regional averages. While vehicle occupants have a 20% share of 3-star or better roads, motorcyclists remain particularly vulnerable, with only 11% of roads meeting this safety standard” [12]. So, according to recent reports, it is visible that motorcycle accidents are one of the main slayers of lives during transport. There are existing laws like driving bikes at moderate speed, using helmets, and using the right lanes, etc. According to The Motor Vehicles Ordinance, 1983 (Ordinance No. LV of 1983) [22nd September, 1983], Chapter VII: “(1) No driver of a two-wheeled motorcycle shall carry more than one person in addition to himself on the cycle and no such person shall be carried otherwise

than sitting on a proper seat securely fixed to the cycle behind the driver's seat.

(2) No driver of a two-wheeled motorcycle shall drive any motorcycle unless he wears a helmet of the prescribed type or carry any person on a motorcycle unless that person wears a helmet of the prescribed type” [13].

Recognizing the urgency of the issue, the World Bank and BRAC, in collaboration with the government, hosted the Standard Motorcycle Helmet Advocacy Event on Sunday, 23 February 2025, at Hotel InterContinental, Dhaka, reports a press release with the theme “Helmet should be a life-saving device, not just a legal requirement” [14]. Even though laws exist and seminars to raise awareness are being held, it is tough to motivate the mass people to abide by laws in such an overpopulated country like Bangladesh. It also has poor infrastructure. Therefore, precise monitoring and digitization of law enforcement are inevitable.

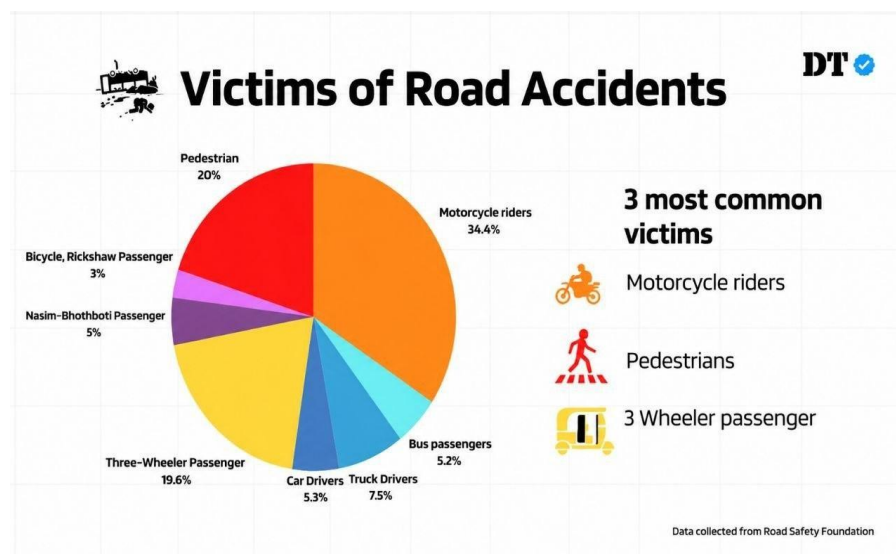


Fig. 2.1 Percentage Distribution of Road Accident Victims in 2024

Figure 2.1 shows the dreadful reality of road accidents through different vehicles in Bangladesh. The orange portion shows motorcycle riders are the worst victim of this[15].

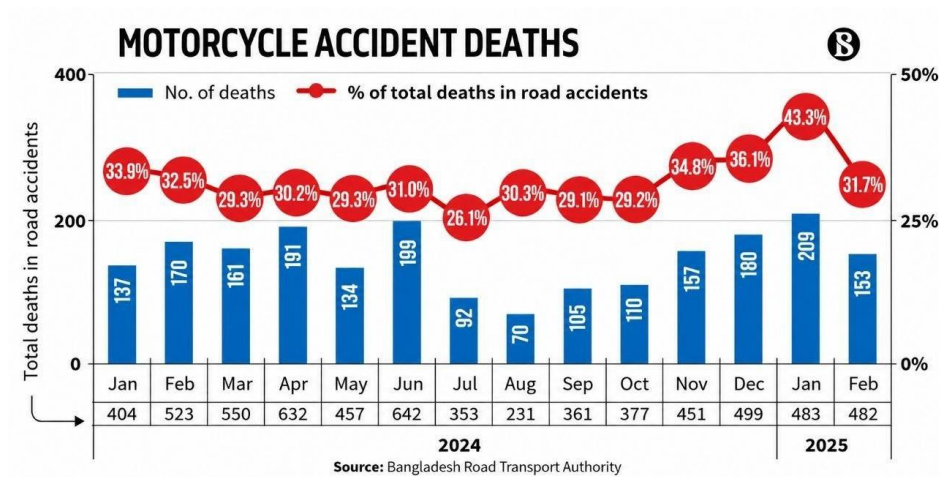


Fig. 2.2 Monthly Analysis of Motorcycle Accident Deaths.

Figure 2.2 shows that what percentage of people have died in motorcycle accidents throughout the years of 2024 and 2025[16].



Fig. 2.3 Comparison of Road Accident Statistics in Bangladesh between 2024 and 2025

A report published in “Prothom Alo” on 10th June,2026 shows the increasing rate of accidents, fatalities and injured over the years by heading “Road accidents killed over 7,300 people in 2025, motorcycles involved in 40pc cases: Road Safety Foundation”[17].

Motorcycle accident is a major concern in Bangladesh and it occurs for many reasons such as-

- **Overspeed** – Young bike riders tend to drive recklessly, make competition on road.
- **Avoiding helmets** – Although being a safety tool, many people avoid it out of discomfort.
- **Using wrong lane**- The roads are narrow in Bangladesh comparing other developed countries. So, any wrong way movement can cause severe injury even death.
- **Overloading** - Riding more than two people in a single two-wheel vehicle can create imbalance in the vehicle.
- **Poor infrastructure** - In rainy days the road becomes muddy or slippery.
- **Un-licensed Drivers**- Not knowing driving can cause severe accident.

2.3 OBJECT DETECTION MODELS IN TRAFFIC SURVEILLANCE

Object detection forms the foundation of vision-based traffic monitoring systems. Detection approaches are generally divided into two-stage detectors, such as Faster R-CNN, which first propose candidate regions and then classify them, and single-stage detectors, such as the YOLO family, which process the entire image in a single pass. Although two-stage detectors can achieve higher accuracy for small objects, their computational requirements make them less suitable for real-time applications.

The YOLO (You Only Look Once) models have evolved significantly since the first version. YOLOv5 introduced the CSPDarknet backbone and Mosaic Augmentation, providing a balance between speed and accuracy. YOLOv8, released by Ultralytics in 2023, further improved feature extraction, anchor-free detection, and a unified framework that supports object detection, segmentation, and pose estimation. This version is particularly suitable for traffic surveillance because it achieves a good trade-off between inference speed and detection accuracy [3]. On September, 2024 YOLOv11 was released.

2.4 HELMET VIOLATION DETECTION

Sowmya et al. [18] created an automated traffic enforcement system using YOLOv11 for helmet detection and OCR to read license plates. Their system achieved 95.2% mAP for helmets and 98% accuracy for plates. Comparing different YOLO versions, they found that YOLOv11x performed best, reaching 54.7 mAP_{50–95} while using fewer parameters and less computation than YOLOv5x and YOLOv8x, showing it is efficient and accurate for real-time traffic violation detection.

Elahi et al. [19] investigated real-time monitoring of traffic violations using a vision-based approach. The study demonstrated that single-stage object detection frameworks are capable of detecting multiple traffic-related objects, such as vehicles and helmets, in a single frame with high speed and accuracy. The authors highlighted that these systems can maintain performance even under low-quality video conditions, making them suitable for deployment with CCTV infrastructure in urban environments. This work supports the selection of YOLOv11 for our study, as it similarly balances detection speed and precision, efficiently identifies helmets and license plates, and enables real-time violation monitoring in practical roadside scenarios.

2.5 WRONG LANE VIOLATION DETECTION

Wrong-way driving is a major contributor to road accidents and traffic congestion. Traditional detection methods face challenges such as sensor errors and occlusions. To address this, Rahman et al. [20] proposed a real-time system using YOLO for vehicle detection and centroid tracking to determine vehicle direction by comparing centroid positions across frames. Tested on real-world video from Chittagong, Bangladesh, the system achieved nearly 100% accuracy, demonstrating robustness and effectiveness under diverse conditions.

2.6 VEHICLE SPEED ESTIMATION

Rahman et al. et al. [21] proposed an IoT-based system for vehicle speed monitoring and control to reduce road accidents in Bangladesh. The system detects speeding vehicles using sensors, microcontrollers and wireless modules. Sends immediate alerts to drivers and traffic authorities. By automating monitoring and providing quick feedback, it reduces dependence on manual enforcement and helps prevent accidents. This approach supports real-time traffic safety and complements AI-based systems using YOLO for violation detection and CRNN for license plate recognition.

Monica et al. [22] developed a system for detecting overspeed vehicles on highways, emphasizing real-time alerts to traffic authorities for improved road safety. The approach employs infrared (IR) sensors positioned at specific points to measure vehicle speed from travel time, without relying on pre-existing datasets. As vehicles pass between the transmitter and receiver, the system calculates speed and triggers a buzzer alert if the speed exceeds the limit, facilitating immediate enforcement of traffic regulations.

Yalabaka et al. [23] developed a CNN-based lane detection system for autonomous vehicles and driver assistance. Using over 5,000 road images under

varied lighting and weather conditions, the model extracted high-level features through multiple convolution, pooling and fully connected layers. Trained on 80% of the dataset and tested on 20%, it achieved 95.64% accuracy, outperforming Random Forest, SVM and other methods, demonstrating CNNs' effectiveness for real-time lane detection and traffic monitoring.

2.7 LICENSE PLATE DETECTION AND RECOGNITION

The paper by Zhao et al. [24] proposes a hybrid system for real-time license plate detection and recognition using YOLO for detecting plates and CRNN for reading the characters. The system is designed to work under different lighting conditions and handle blurred or partially blocked plates. It was trained on a dataset of annotated vehicle images and achieved high accuracy for both detection and recognition. The framework can process video in real time and store recognized plates in structured logs, making it suitable for traffic monitoring, toll management, and urban surveillance applications.

2.8 SUMMARY AND IMPLICATIONS

The literature demonstrates a clear trend toward automated traffic monitoring systems using computer vision and deep learning, driven by the limitations of manual enforcement in urban environments. Early works relied on YOLO variants for vehicle detection and single-stage object detection tasks, proving the efficiency and scalability of CNN-based methods in real-time traffic surveillance (Franklin et al. [6], Pavithra et al. [7]).

Object detection models, especially YOLOv8, have evolved to balance speed and accuracy, making them highly suitable for traffic monitoring tasks, including vehicle, helmet, and license plate detection [3]. Helmet compliance detection has been addressed by Sowmya et al. [18], demonstrating that YOLOv11 can efficiently automate helmet monitoring while providing real-time alerts. Wronglane detection and lane violation systems using YOLO and centroid

tracking have shown near-perfect accuracy under real-world conditions (Rahman et al. [25]).

Similarly, vehicle speed estimation via IR sensors and lane detection using CNNs provide effective mechanisms for detecting overspeed and guiding autonomous traffic monitoring. For license plate recognition, hybrid YOLO-CRNN frameworks handle challenging lighting, occlusion and blurred plates, achieving high real-time performance [24].

YOLO-style single-stage detectors treat object detection as a direct regression problem across the entire image, enabling high frame rates with acceptable accuracy. Previous studies have shown that YOLOv3, YOLOv5, YOLOv8, YOLOv11 can detect vehicles, helmets and other traffic objects in real time and are widely used in end-to-end traffic enforcement systems such as SafeStreets. In the proposed system, YOLO-family models are used to detect helmets and non-helmeted riders, motorcycle license plates, and riders and motorcycles from a COCO-trained model to estimate the number of riders per vehicle.

Convolutional Recurrent Neural Networks (CRNN) combine CNN feature extraction with recurrent layers and CTC loss to recognize variable-length character sequences directly from images. In this system, detected plate images are upscaled and pre-processed, then passed through a CRNN trained on Bangla and alphanumeric characters. CTC greedy decoding is applied to produce the recognized plate numbers.

The proposed system can be divided into three main stages:

- **Perception:** YOLO-based detection of helmets, riders, motorcycles, and license plates.
- **Reasoning:** Determination of traffic violations, such as no helmet or more than two riders per motorcycle.

- **Identification:** Localization and recognition of license plate text using CRNN to identify offending vehicles.

Chapter 3: MATERIALS AND METHODOLOGY

3.1 DEEP LEARNING (DL)

Machine learning is a form of artificial intelligence or AI. It helps a system to learn data without active programming. Thus, a system can learn pattern and recognize it. Deep learning is a special form of machine learning. A neural network is developed through deep learning. There are three general layers in deep learning model. Input layer receives data. Hidden layers learn the features. Output layers give output based on learning. A neural network is used to recognize image, speech etc.

YOLOv11 is an object detecting deep learning system. It can easily analyze day to day data of traffic scenario. It can automatically detect if any motorcycle rider disobeys the traffic rules.

3.1.1 YOLOv11: An Advanced Real-Time Object Detection Model

YOLOv11 (You Only Look Once Version 11) is an advanced object detection model designed for fast and accurate real-time object recognition. It uses a singlestage detection approach, allowing object localization and classification to be performed simultaneously.

Compared to earlier versions, YOLOv11 provides improved feature extraction and computational efficiency, enabling more accurate detection of small and overlapping objects. Due to its speed, accuracy, and lightweight architecture, YOLOv11 is well suited for intelligent traffic monitoring

applications such as detecting motorcycles, riders, helmets, and license plates in real time.

3.1.1.1 Key Features of YOLOv11 The

main features of YOLOv11 are:

- Real-time object detection.
- High detection accuracy.
- Effective detection of small objects.
- Efficient feature extraction.
- Lightweight architecture.
- Multi-object detection capability.
- Robust performance under varying traffic and lighting conditions.

3.1.1.2 Model Variations

YOLOv11 is available in different sizes to meet various speed and accuracy requirements:

- **YOLOv11n (Nano):** Fastest and smallest model.
- **YOLOv11s (Small):** Balanced speed and accuracy.
- **YOLOv11m (Medium):** Higher accuracy with moderate complexity.
- **YOLOv11l (Large):** Improved detection performance.
- **YOLOv11x (Extra Large):** Highest accuracy but greater computational requirements.

3.1.1.3 Application of Road Safety Context in Bangladesh

In Bangladesh, where motorcycle related accidents and traffic violations are common, YOLOv11 can contribute to road safety by:

- Detecting helmet violations and multiple riders on motorcycles.
- Identifying wrong-way movement and overspeed motorcycles.
- Monitoring motorcycle behavior in real time.
- Assisting traffic authorities through automatic violation detection.
- Supporting accident prevention and traffic law enforcement.

3.1.1.4 Justification for Selecting YOLOv11

YOLOv11 was selected as the object detection model for this research because it provides a good balance between speed, accuracy, and efficiency. The main reasons are:

- **Real-Time Performance:** It processes video frames quickly, making it suitable for traffic monitoring.
- **High Detection Accuracy:** It accurately detects motorcycles, riders, helmets, and license plates, even in busy traffic scenes.
- **Efficient Resource Usage:** Its lightweight design allows it to run efficiently without requiring high computational power.
- **Small Object Detection:** It performs well in detecting small objects such as helmets and license plates.
- **Easy Implementation:** It supports different model variants and can be easily integrated into Python-based computer vision applications.

3.1.2 CRNN MODEL

A CRNN is designed for sequence recognition in images, such as text recognition (OCR), license plate reading, handwriting recognition, and speech spectrogram analysis. It combines the strengths of CNNs for feature extraction with RNNs for sequence modelling.

3.1.2.1 CRNN Architecture

- Convolutional Layers (CNN):
 - Extract spatial features from images.
 - Detect edges, shapes, and patterns.
 - Reduce dimensionality while preserving important information.
- Recurrent Layers (RNN / BiLSTM):
 - Process sequential data.
 - Capture temporal or sequential dependencies (e.g., order of characters).
 - Bidirectional LSTM (BiLSTM) is commonly used to capture both forward and backward context.
- CTC (Connectionist Temporal Classification) Layer:
 - Maps the variable-length RNN output to a sequence of labels.
 - Handles alignment without pre-segmented input (important for text with variable length).
- Output Layer:
 - Produces a sequence of probabilities over the character set.
 - Decoding methods (like greedy or beam search) convert this to actual text.

3.1.2.2 Key Feature of CRNN

- CRNN can be trained from input images directly to output sequences without manual segmentation of characters.
- CNN layers extract spatial features from images. RNN layers handle sequential dependencies, making it suitable for variable-length sequences.
- Can process inputs of varying widths and output sequences of different lengths, which is crucial for license plates with different numbers of characters.
- Eliminates the need for pre-aligned labels.
- Robust to Noise and Distortion
- Can be trained on different character sets, including alphanumeric, Bangla, or multilingual plates.
- Lightweight CRNN architectures can process images quickly.

3.2 LIBRARY DESCRIPTION

The libraries used in the project are:

- **Ultralytics.YOLO:** The interface for YOLOv8 models, allowing loading of pre-trained weights (e.g., yolov8n.pt), performing object detection and segmentation, and obtaining bounding boxes, class labels, and confidence scores.
- **cv2 (OpenCV):** OpenCV handles video and image processing tasks, including reading frames, drawing bounding boxes, and displaying detection results. It is central for preprocessing, visualization, and working with real-time video streams.
- **JSON:** Provides serialization of Python objects into JSON format for

API communication or structured storage.

- **Time:** Used to measure processing times, frame delays, and timestamps for detections, which helps evaluate the real-time performance of YOLOv8 inference.
- **Math:** The standard Python math library is used for computations such as distances between objects, angle calculations, and logarithmic operations, which assist in post-processing detection results.
- **NumPy:** Provides efficient array operations and numerical computations. It is essential for reshaping images, creating masks, and performing transformations in the computer vision workflow.
- **matplotlib.pyplot:** Used to plot graphs, display intermediate results, and visualize metrics such as detection confidence over time. This is useful for analyzing performance and reporting results.
- **Imutils:** A utility library built on top of OpenCV that simplifies common image processing operations, such as resizing, rotating or changing image aspect ratios.
- **pymongo.MongoClient:** Connects to a MongoDB database to store detection logs, license plate numbers, timestamps, and violation information. This allows historical tracking and analysis of road safety metrics.
- **Datetime:** Captures the date and time of each detection or violation. This information is stored in the database to enable auditing, record keeping and forensic analysis of traffic violations.
- **PyTorch:** Used to build and train neural networks using Python.
- **Requests:** A Python library for making HTTP requests to external APIs

or services. In this project, it is used for triggering alerts to the traffic monitoring dashboard, notifying authorities or police and potentially retrieving external information such as road or weather condition.

3.3 METHODOLOGY

The proposed scheme is designed to automatically track motorcycle rules violation instantly by utilizing the YOLOv11 object detection model. As shown in Fig. 3.2, the system receives video footage through roadside cameras as its input. Captured video is divided into frames, which are then processed by YOLOv11 to detect motorcycles, riders, helmets, and license plates.

The system continuously tracks motorcycles and analyzes their movement to identify different violations, including riding without a helmet, carrying more than two persons, wrong-way movement, and overspeed. Once any traffic rule is disobeyed, the scheme utilizes image and video evidence and extracts characters of the number plate number of that offending motorcycle.

The number plate information is then processed to recognize the characters, and an automated notification module generates alerts containing the violation details. These alerts can be sent to the motorcycle owner and the relevant authorities, while the violation information is stored for future reference. By decreasing dependency of manual effort, the proposed scheme can simultaneously detect multiple violations and improve the efficiency and accuracy of traffic law enforcement.

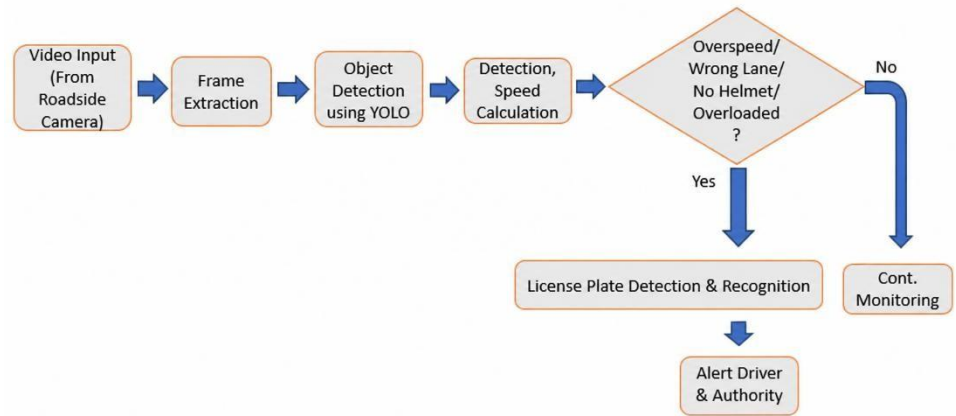


Fig. 3.1 Block diagram of the proposed ITS.

3.3.1 Speed and Wrong Lane Detection

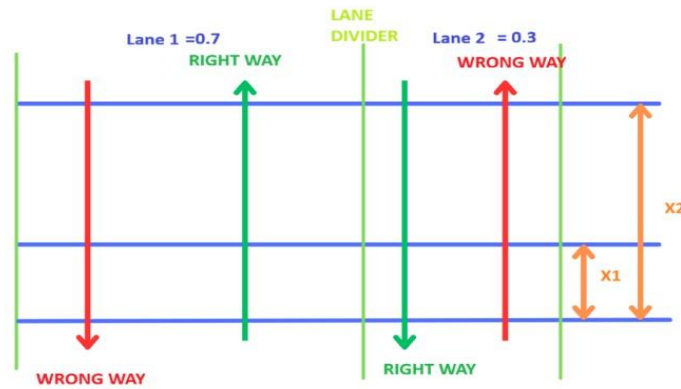


Fig. 3.2 Method of Speed and Wrong Lane Detection.

In this system, depicted in figure 3.3, the camera is placed at a fixed position and used as a reference point to observe the movement of vehicles. Two virtual lines are drawn on the video frame at specific positions so that the movement of each vehicle can be measured as it passes through the monitored area. As shown in Fig. 3.3, the road is divided into two lanes by a lane divider, and the virtual lines are used to detect both vehicle speed and lane direction. Here X_1 is the distance between the camera and first line and X_2 is the difference between the camera and second line. The whole setup works on the idea of space

between pixels. When a vehicle crosses the first line, the system starts counting the time and assigns a unique ID to that vehicle. Time count stops when the bike reaches to other line. The speed of the bike is then measured using Equation (1).

$$V = \frac{X_2 - X_1}{T} \dots (1)$$

Here, V is the speed of the vehicle with its ID, and T is the time that was counted during travel. To detect wrong-way movement, the system checks the order in which the vehicle crosses the lines. If the vehicle crosses the lines in the expected direction, it is considered to be moving in the correct lane direction.

3.3.2 Dataset

3.3.2.1 Dataset Used to train YOLOv11n

The dataset for detecting helmets, no-helmet riders and license plates using YOLOv11n was collected from Roboflow and contains 1,533 images of road conditions. Each image has an associated label file specifying the object class and normalized bounding box coordinates (class, center x, center y, width, height). These annotations allow the model to learn both the category and location of helmets, riders and license plates. All images are 640×640 pixels.

3.3.2.2 Custom Dataset for CRNN Model

A custom dataset was created to train the CRNN model for recognizing Bengali license plates from roadside videos. Since no public dataset existed, videos were recorded across various locations in Bangladesh, capturing vehicles under different conditions. License plates were cropped from these videos and stored in separate folders to reflect real-world traffic scenarios.

A total of 1,089 images were prepared for the project: 1012 for training, 114 for validation, and 87 for testing, ensuring a realistic distribution of plate styles, lighting conditions and occlusions. A charset.txt file was created containing all possible characters present in the license plates. This setup provides a comprehensive learning spectrum for the CRNN model to generalize well across

real-world data.

Each cropped license plate image was labeled by specifying the image path and the corresponding text for training the CRNN. The labels indicate the exact sequence of characters for each license plate, facilitating sequence-based recognition. The dataset was organized into 83% training, 10% validation and 7% test subsets, reflecting common practices for machine learning pipelines to ensure model robustness and evaluation accuracy. This annotation process allows the CRNN to learn both the spatial features of characters through convolutional layers and the sequence dependencies through recurrent layers with CTC loss. The final dataset is compatible with the CRNN training framework and can be directly used for sequence recognition in real-time traffic monitoring applications.

3.3.3 Multiple Rider Detection

A pretrained YOLOv11n model trained on the COCO dataset is used to detect generic object classes, including person and two-wheeler categories directly from each frame. All detected person boxes and vehicle boxes are separated into two groups. For each detected vehicle, the IoU is calculated between its bounding box and every detected person box in the frame, using Eq. (2) :

$$IoU = \frac{Area(B_p \cap B_v)}{Area(B_p \cup B_v)} \dots\dots(2)$$

where B_p and B_v denote the bounding boxes of a person and a vehicle, respectively. A person is considered a rider of that vehicle if the IoU value exceeds a threshold of 0.1. The total number of riders associated with a vehicle is obtained by counting all person boxes that satisfy this overlap condition. If the resulting count exceeds two, the vehicle is flagged as a multi-rider violation.

3.3.4 Message Generation and Delivery

After any confirmed traffic violation, the license plate is recognized, then it calls the Flask endpoint with JSON containing the plate, violation type, time, and any mapped driver phone number. Flask API receives confirmed violation events from detection pipeline, stores them in MongoDB, and then uses Twilio to actually send SMS alerts.

This code is a Flask-based SMS violation API that receives violation data, looks up a vehicle, calculates a fine and sends an SMS using Twilio, while keeping everything in memory.

Chapter 4: RESULTS

For the proposed ITS, a video stream of a main road in Dhaka was used. It was able to detect different vehicles and object on the road successfully and identified the direction of their movement simultaneously for two lanes.



Fig. 4.1 Types of vehicles and wrong way detection.

Figure 4.1 shows how wrong-way and right-way movements are detected. By considering two reference lines, when a vehicle enters the zone, its movement is measured using pixel distance. Based on this movement, the system can identify whether the vehicle is traveling in the wrong direction or the correct direction.

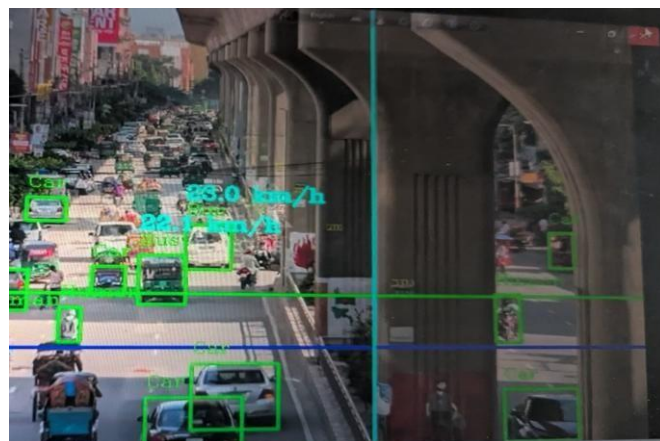


Fig. 4.2 Measurement of velocity when a vehicle crosses both reference lines.

Figure 4.2 shows the speed measurement process using the pixel distance between two reference lines and the time taken by a vehicle to cross them. By calculating the distance traveled and the corresponding crossing time, the vehicle's speed can be determined.



Fig. 4.3 Numberplate detection of vehicle including tracking ID

In figure 4.3, the numberplate was tracked with ID: 1 and a single numberplate was recognized for this ID.



Fig. 4.4 Number of riders and helmet detection

In figure 4.3 the YOLOv11n model is detecting if the riders are wearing helmet or not. Also it is showing number of riders on the bike. The helmet violation is marked with red boundary box which has confidence score over 0.6.

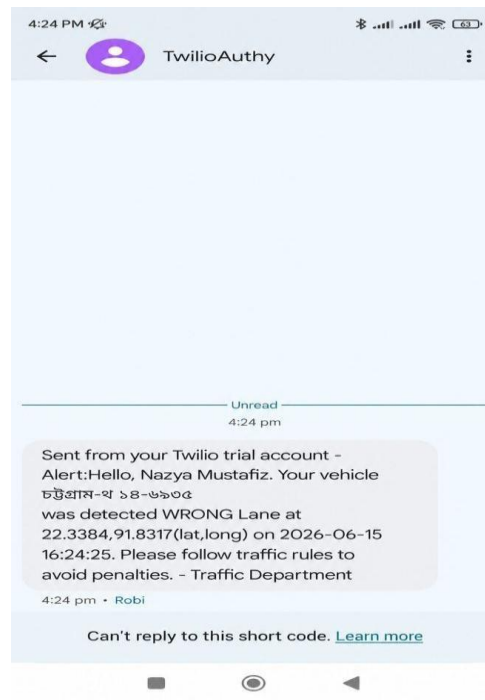


Fig. 4.5 Automated sms system after violation is detected.

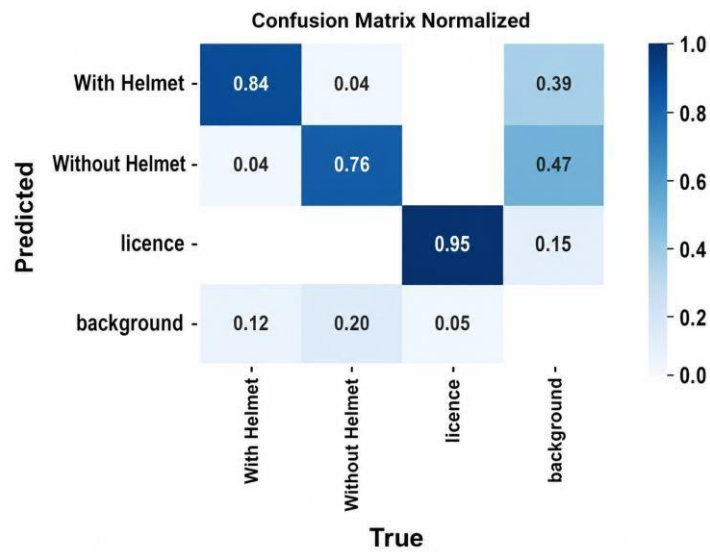
So Basically, Figure 4.1 shows the right way and wrong way movement for s particular vehicle in two lane system. Figure 4.2 shows velocity of a vehicle after the vehicle passes two reference lines. Figure 4.3 shows two riders were detected and none of them wear a Helmet, which is the violation of traffic rule. Figure 4.4 showed an alert to the owner of violated vehicle.

Chapter 5: ANALYSIS

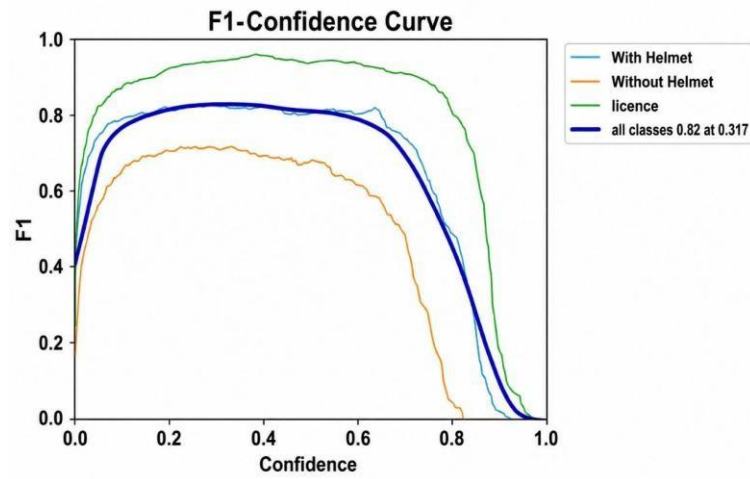
The choice of YOLOv11 over older models like YOLOv8 and YOLOv5 is justified by its improved object detection accuracy and faster training. YOLOv11 can identify vehicles of different types under various conditions more accurately and processes images in real time, making it well-suited for dynamic traffic monitoring where immediate action is required.

Compared to prior work, this system offers several improvements. Earlier studies focused on specific tasks, such as wrong-way detection or speed measurement, but often lacked real-time integration, automation, or alert features. For instance, Rahman et al. developed a YOLO-based wrong-way detection system but did not include instant notifications to authorities, and Murshed et al.'s ultrasonic speed estimation system had low precision and relied heavily on sensors. In contrast, the proposed system uses existing road surveillance infrastructure, making it cost-effective, scalable, and ready for deployment in developing countries like Bangladesh.

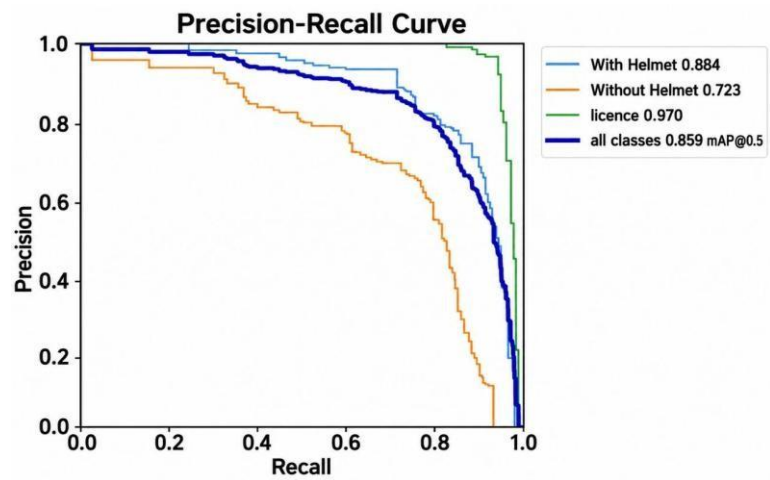
Although the system performs well, it has certain limitations. The current dataset used for training the license plate recognition model may not include enough variation in lighting or plate designs, especially for Bengali plates. The recall rate may be relatively low, meaning that some vehicles could go undetected under specific conditions. Additionally, environmental factors such as fog, poor lighting, and occlusions can reduce detection accuracy.



(a)



(b)



(c)

Fig. 5.1 (a)Confusion matrix , (b) F1 confidence curve, (c) Precision -recall curve.

The system was evaluated using per-class metrics, including a normalized confusion matrix Fig. 5.1(a), F1-confidence curves, and precision-recall analysis. In the confusion matrix, the “With Helmet” and “Without Helmet” classes achieved 84% and 76% correct classification, respectively, while the “license” class performed best at 95%. Some confusion exists between the background and helmet classes (0.39 and 0.47), indicating occasional false detections of non-object regions.

The F1-confidence curve Fig 5.1(b) shows that the “license” class maintains the highest and most stable F1 score, peaking near 0.95, whereas the “Without Helmet” class is the weakest and most variable, peaking around 0.71. The combined F1 curve for all classes reaches a maximum of 0.82 at a confidence threshold of 0.317, suggesting this threshold optimally balances precision and recall.

The precision-recall analysis fig 5.1(c) highlights the trade-off between precision and recall at varying confidence levels. The “license” class achieved the highest average precision (0.970), followed by “With Helmet” (0.884), while “Without Helmet” lagged at 0.723, reflecting more false positives and missed detections. The overall mAP@0.5 across all classes is 0.859, demonstrating strong detection performance for real-time traffic monitoring.

Beyond the detection metrics showed in Fig. 5.1, we also tested 15 different traffic video from the road of Chattogram Highway. The videos included over 100 different Bangladeshi vehicles, each given a fixed tracking ID. It means we tested plate recognition on real time data not from cropped plate.

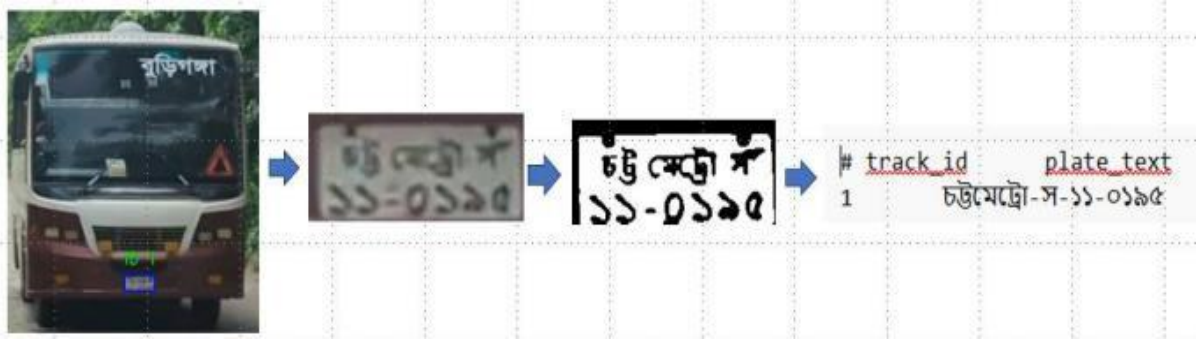


Fig. 5.2 Detection and Extraction of Bangla Characters from Nameplates.

Fig. 5.2 shows the four steps for one of the tracked vehicles correctly, Those four steps were detecting, cropping, binarizing and then finally reading the characters using custom CRNN Model. But Bangla plates are a bit harder to read than Latin or English plates because some Bangla characters look almost identical. Like 1,3,5,9,0 etc have closed loop in one side or both sides. A real example was seen in our test: The Bangla digit for five was sometimes misread as the Bangla digit for zero. At low resolution or after binarization, or even in smudged plate, the curve of those two digits read identical in two cases.

Error happened most often when the plate was:

- Inclined - tilting the angle of camera distorts curved Bangla strokes more than it can distort the straighter strokes of Latin or English letters.
- Smudged - dirt tends to break exact strokes, from which we understand two characters are apart. Both effects actually explained why the Levenshtein-distance matching score was lower for inclined or smudged plates during testing.

Class	Correct Detection	AP@0.5	Peak F1 Score	Notes
With Helmet	84%	0.884	—	Confused with background in ~39% of misclassified cases
Without Helmet	76%	0.723	~0.71	Weakest and most variable F1 curve
License Plate	95%	0.970	~0.95	Highest and most stable F1 curve
All Classes (combined)	—	mAP=0.859	0.82	Optimal confidence threshold = 0.317

Fig. 5.3 Comparison of Result for Different Classes.

From the figure we see that detection accuracy without helmet is a bit lower. It is due to human faces are not identical, it varies a lot for different shape and hairstyles. But shape of helmet is smooth to detect. That's why, "without helmet" tends to a lower accuracy of 76%. Overall mAP was appeared as 85.9%.

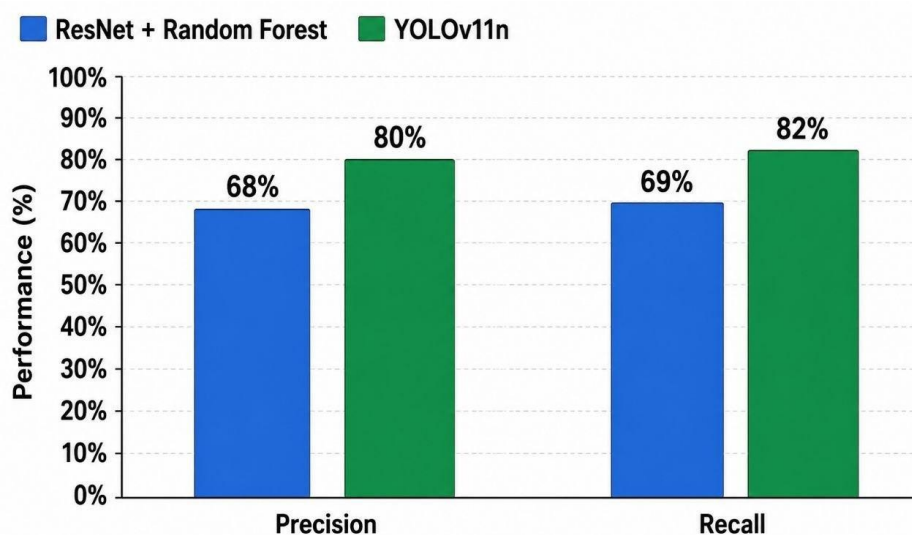


Fig. 5.4 Performance analysis between YOLO v11n and Resnet50+RF.

As depicted in Figure 4.7, the performance of the ResNet50+Random Forest model was compared with YOLOv11n to determine the better option for helmet violation detection. The ResNet50+RF model reached an overall accuracy of 68.65%, with weighted precision and recall of 0.68 and 0.69, respectively. Its lower performance is mainly due to its two-stage process, where object regions are first detected and cropped before classification, which increases the chance of errors. It also had difficulty distinguishing between Helmet and No-Helmet classes, as these look very similar. In contrast, YOLOv11n achieved a precision of 80.3%, recall of 82.4%, and mAP50 of 84.9% after 30 training epochs, because it performs detection and classification in a single step. This simpler, end-to-end design reduces errors and allows the model to learn spatial and context information more effectively. Based on these results, YOLOv11n was chosen as the final model due to its higher accuracy, better generalization, and suitability for real-time use.

Chapter 6: CONCLUSIONS

7.1 GENERAL

Motorcycle-related road accidents have become a serious problem in Bangladesh, resulting in significant loss of lives and property every year. Many of these accidents are associated with common traffic violations such as riding without a helmet, exceeding speed limits, carrying more passengers than allowed, and using the wrong lane. Ensuring compliance with traffic regulations through manual monitoring is often difficult due to limited resources and the large volume of traffic.

This research explores the use of artificial intelligence to automatically identify these violations and assist traffic management authorities. The proposed system analyses road traffic video footage, detects violations committed by motorcycle riders, recognizes vehicle license plates, and sends instant notifications when an offense is identified. By combining automated monitoring with real-time reporting, the study seeks to provide a practical solution that can support safer roads, improve enforcement efficiency, and reduce dependence on continuous human observation. The overall goal is to demonstrate how intelligent technologies can contribute to an efficient and modernized vehicle management in Bangladesh.

7.2 KEY FINDINGS

The study intended to address the issue of motorcycle accidents in Bangladesh, in addition to general vehicle-related violations. It focuses on four common motorcycle traffic violations: not wearing helmets, overspeed, overloading, and driving in the wrong lane. By developing an Artificial Intelligence Algorithm, the system can monitor, detect, and report disobeying traffic rules in real life.

To achieve this, YOLOv11 was used as the primary object detection model due to its high precision, along with an automated deep learning model for license plate detection and recognition. After detecting a violation, Twilio's SMS gateway was used to alert drivers and authorities. Through testing on real-life video footage, the system achieved 95% accuracy in number plate detection and 97% accuracy in helmet detection, confirming its technical effectiveness. Compared to previous solutions, the system successfully presents a specialized, real-time, fully automated, and scalable traffic monitoring framework. It can help developing countries like Bangladesh overcome infrastructure challenges and reduce errors caused by human intervention.

7.3 LIMITATION OF THE STUDY

Although the system can identify different types of vehicles, traffic signal violation detection is not included in the current system. The system was mainly tested on vehicles in Bangladesh. It cannot detect license plates from all languages in the world, as it is specifically designed for Bangla license plates. Furthermore, the detection system requires a minimum level of lighting to operate effectively. The national vehicle registration database is not integrated into the system and it cannot detect seat belt usage. Lastly, the system does not include privacy protection mechanisms such as data encryption and

anonymization. Addressing these limitations will be crucial for to improve execution of this scheme, thus enhancing road safety.

7.4 RECOMMENDATION FOR FURTHER STUDY

In the near future, Bangla nameplate detection and extraction can be significantly improved by training the model with a larger dataset that contains real-life traffic scenarios under various conditions. To enhance performance, different versions of YOLO can be used and compared to determine the bestperforming model. Hybrid approaches can also be explored by integrating pattern recognition techniques with tools such as MATLAB. Additionally, improving the system's processing speed can make it more efficient. The inclusion of database of registered motorcycle would enable verification of violated traffic automatically and facilitate the identification of unregistered vehicles. Furthermore, future development should focus on improving performance under extreme low-light conditions and in detecting traffic signal violations.

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